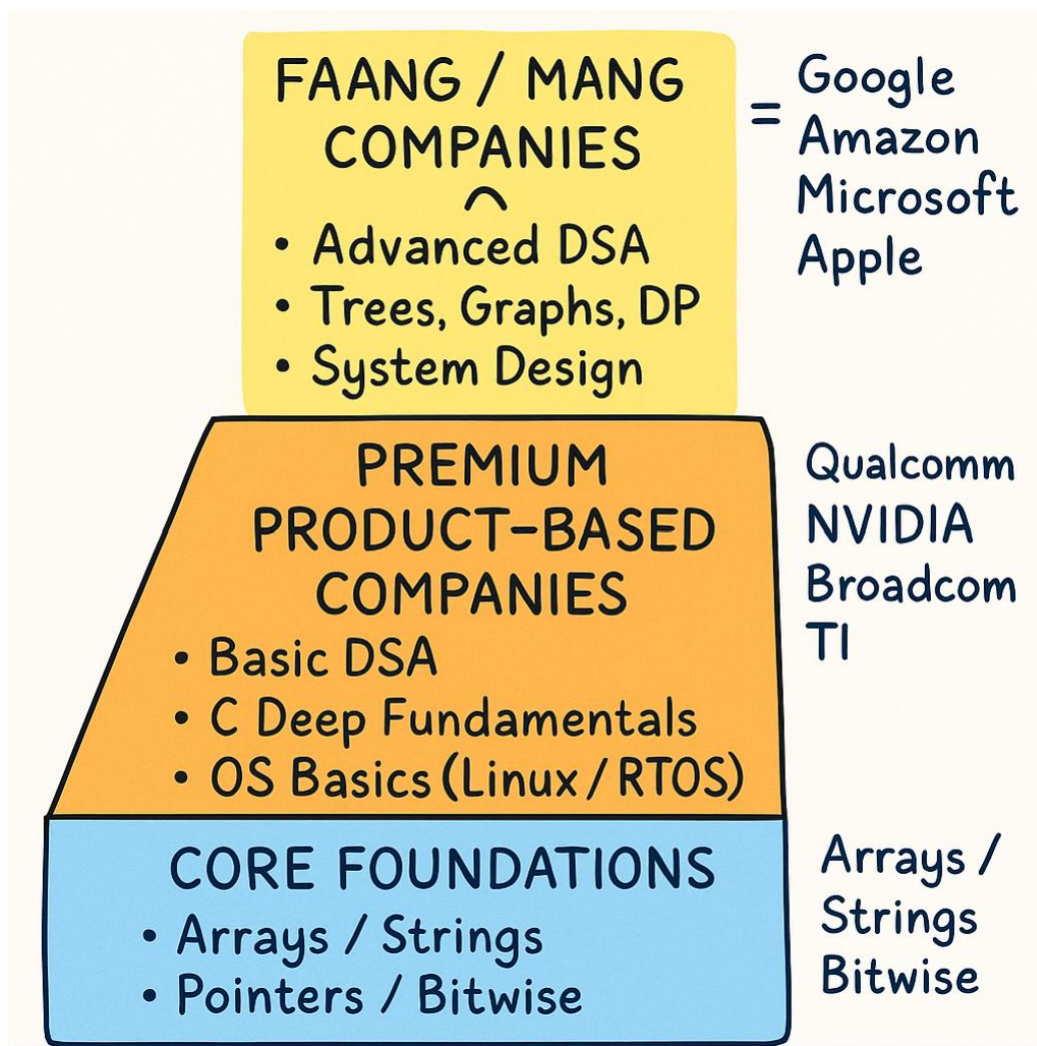




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## ■ EMBEDDED SYSTEM INTERVIEW PREPARATION — PART 1

*(For Premium Product-Based Companies — Not MAANG/FANG)*

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### ★ Introduction

If you're preparing for **premium product-based embedded system companies** (e.g., high-quality startups, consumer electronics firms, automotive product companies, networking companies, semiconductor companies, etc.), your interview preparation should be **focused, practical, and core-skill oriented**.

These companies do **NOT** expect MAANG-level DSA.

They want strong fundamentals in **C programming, OS concepts, embedded concepts, and debugging skills**.

This document explains **exactly what to study and how to prepare**.

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## **PART A — Core Areas You MUST Master**

Premium product companies mainly evaluate four things:

1. **Strong Command Over C Programming**
2. **Solid Understanding of OS Fundamentals (Linux OR RTOS)**
3. **Basic–Intermediate Data Structures (But NOT Leetcode Hard)**
4. **Embedded Concepts + Debugging Mindset**

Let's go step-by-step.

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### ◆ **1. C Programming Fundamentals (Most Important)**

C is the heart of embedded interviews.

You don't need competitive programming.

You need **system-level, memory-level C**.

#### ✔ **Topics to Master**

- Pointer basics
- Pointer-to-pointer
- Pointer arithmetic
- Arrays & strings
- Function pointers
- Const, volatile, restrict
- Structures, unions, bitfields
- Dynamic memory (malloc/free) — why avoided in embedded
- Memory layout (stack, heap, BSS, data, text)
- Storage classes (auto/static/register/extern)
- Compilation steps, header vs source
- C program execution flow
- Integer promotions & operator precedence

- Common pitfalls and undefined behavior

### ✔ What to practice?

- Reverse a string
- Find length of string
- Print linked list
- Stack/Queue using array
- Remove duplicates from an array
- Bit manipulation problems
- Write your own strcpy, strlen, memcpy, strcat
- Endianness check
- Swap using XOR
- Implement circular buffer
- Simple state machines

This is **enough** for 90% of product-based embedded interviews.

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## ◆ 2. OS Basics (Linux or RTOS)

Companies expect you to know **how the system works**, NOT deep kernel internals.

### ✔ If preparing with Linux:

- What is a process?
- What is a thread?
- Scheduling basics
- Context switching
- Virtual memory basics
- System calls
- Fork/exec concept
- Mutex vs semaphore
- Deadlock basics
- Interrupt/User space difference

- How device drivers abstract hardware
- File descriptors
- Blocking vs non-blocking
- Poll/select basics
- Simple shell commands (ls, ps, top, dmesg)

✓ **If preparing with RTOS:**

- Task creation
- Task switching
- Task states
- Mutex, semaphore, event groups
- Priority & priority inversion
- Tick interrupt
- Delay API (vTaskDelay / osDelay)
- Queues and message passing
- ISR vs task context
- Memory pools/static allocation

You don't need to read Linux kernel book.  
Just **practical OS knowledge**.

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◆ **3. Basic Data Structures (NOT Leetcode DSA)**

Premium product companies do **NOT** expect red-black trees or DP questions.  
They focus **ONLY** on structures used in real embedded code.

✓ **Must-know data structures:**

- **Array**
- **String**
- **Linked List (basic)**
- **Stack (array-based)**
- **Queue (array-based / circular buffer)**

## Why?

Because these are actually used in embedded:

- Linked lists for RTOS queues
- Circular queues for UART buffers
- Stacks for expression parsing & task stacks

### ✔ Practice questions:

- Implement queue using array
- Implement stack
- Insert/delete in linked list
- Reverse a linked list
- Produce-consume using circular buffer
- Basic string handling (uppercase, removal, palindrome)

That's all. Nothing beyond this is required for premium product companies.

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## ◆ 4. Embedded Systems Fundamentals

You must know how hardware interacts with software.

### ✔ Topics:

- GPIO, UART, I2C, SPI basics
- Registers & bit manipulation
- Polling vs interrupt
- Debouncing
- Timers & counters
- Watchdog
- PWM basics
- Memory-mapped I/O
- Basic board bring-up flow
- Reading a datasheet / reference manual basics
- Booting process overview

- Interrupt flow (ISR, vector table)

### ✓ **Debugging Mindset:**

- How to trace a bug
- Understanding stack overflow
- Infinite loops
- Race conditions
- How to use logs
- How to reproduce issues
- Boundary cases
- Timing issues
- Handling NULL pointers & corrupted memory

This is what premium companies value most.

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## **PART B — Interview Strategy & Preparation Plan**

### **Step 1: Master C fundamentals (20–30 days)**

Practice small programs every day.

### **Step 2: Learn OS basics (Linux OR RTOS) (15 days)**

Just basics + practical understanding.

### **Step 3: Solve basic data structure problems (20 days)**

Arrays, strings, linked lists, stack, queue.

### **Step 4: Learn core embedded concepts (20 days)**

Timers, interrupts, protocols, state machines.

### **Step 5: Build 1–2 hands-on mini projects**

Example:

- I2C temperature sensor
- UART communication
- LED + timer + interrupt demo
- Small FreeRTOS program

## Step 6: Prepare for interview patterns

- Round 1: C programming
  - Round 2: Embedded + OS
  - Round 3: Embedded debugging
  - Round 4: System thinking scenario
  - Round 5: Manager/HR
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## PART C — What They Do NOT Expect

These companies **do NOT** require:

- ✗ Dynamic programming
- ✗ Graph algorithms
- ✗ Trees, heaps, tries
- ✗ Leetcode medium/hard
- ✗ Compilers, networking stacks
- ✗ Deep Linux kernel internals

Your preparation should be **focused**, not scattered.





# JOIN THE BOOTCAMP

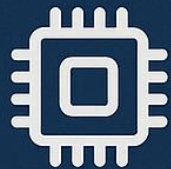
Structured interview prep  
covering:



C



RTOS



Drivers



Mock  
interviews



Resume prep  
guidance



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## FINAL MESSAGE

To clear premium product-based embedded interviews:

**Master C + OS basics + simple DSA + embedded fundamentals.**

No need for MAANG-style DSA.

Practical knowledge > fancy algorithms.

Stay Tuned for Part 2

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